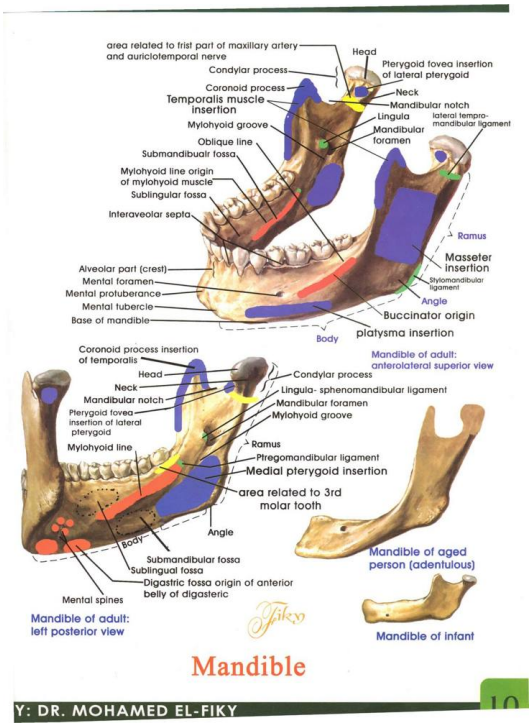
# Mandible

Parts, surfaces, foramina, muscle attachments and nerves of the lower jaw, with how it changes with age.

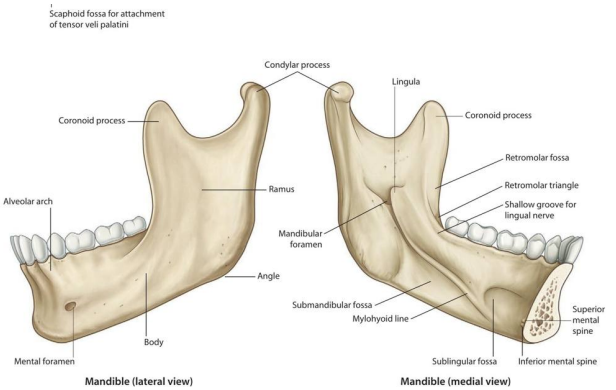
## Parts of the mandible

- **A horseshoe-shaped body** and **two rami**, meeting at the **angle** on each side.
- **Ramus:** ends above in the **coronoid process** (front) and **condylar process** (back), separated by the **mandibular notch**.
- **Body:** an upper **alveolar part** carrying the teeth and a lower **base**.
- **In the midline:** symphysis menti and the mental protuberance, with a mental tubercle on each side.

**Clinical pearl:** The mandible develops around Meckel's cartilage of the first arch but is mostly formed by membranous ossification.




Mandible: adult, infant and aged. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 10

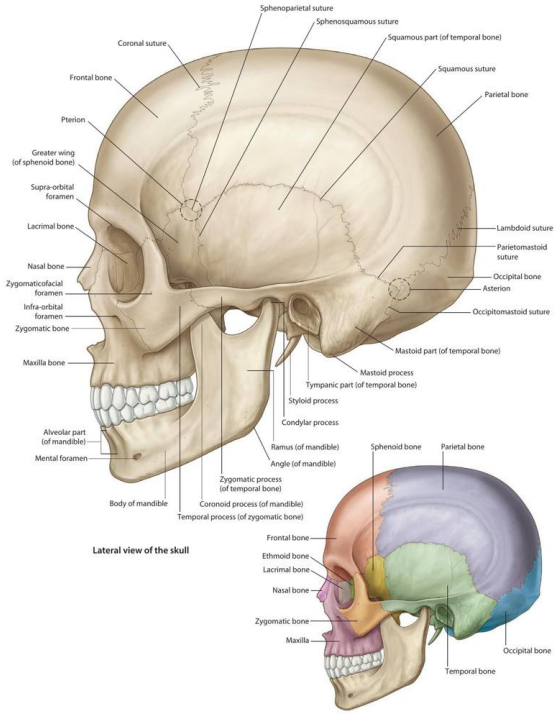



Mandible (lateral and medial views). Gray's Atlas of Anatomy, 2nd ed., p. 572

## Outer surface

- **Mental foramen:** near the apex of the second premolar, midway between the upper and lower borders in the dentate adult. Transmits the mental nerve and vessels.
- **Oblique line:** runs from the anterior border of the ramus onto the body. Gives origin to depressor labii inferioris and depressor anguli oris; buccinator arises beside the molar teeth.
- **Lateral ramus and angle:** masseter.
- **Incisive fossa:** mentalis. **Lower border:** platysma.

**Clinical pearl:** The facial artery crosses the lower border at the front of masseter, where you can feel its pulse.

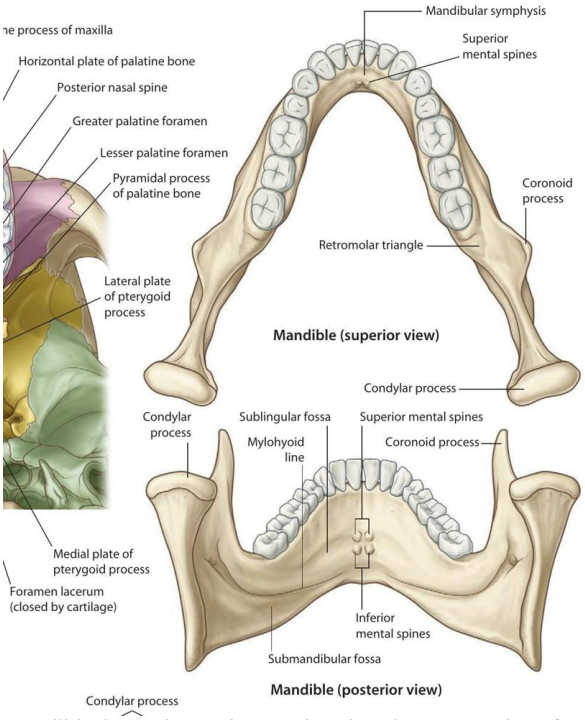



Lateral view of the skull: squamous, tympanic and mastoid parts. Gray's Atlas of Anatomy, 2nd ed., p. 480

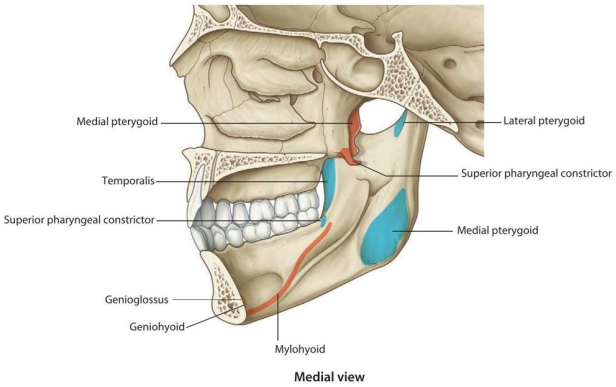
## Inner surface

- **Mandibular foramen:** centre of the medial ramus. Transmits the **inferior alveolar nerve and vessels**.
- **Lingula:** a tongue of bone in front of the foramen. Gives attachment to the **sphenomandibular ligament**.
- **Mylohyoid groove:** below and behind the foramen, for the nerve to mylohyoid and its vessels.
- **Mylohyoid line:** gives origin to mylohyoid. **Sublingual fossa** lies above its front part, **submandibular fossa** below its back part.
- **Superior mental spine:** genioglossus. **Inferior mental spine:** geniohyoid. **Digastric fossa:** anterior belly of digastric.
- **Medial angle:** medial pterygoid. **Behind the last molar:** pterygomandibular raphe and superior constrictor.

**Clinical pearl:** The lingual nerve lies against the bone just below and behind the third molar, so it is at risk during extraction.




Mandible (superior and posterior views). Gray's Atlas of Anatomy, 2nd ed., p. 572

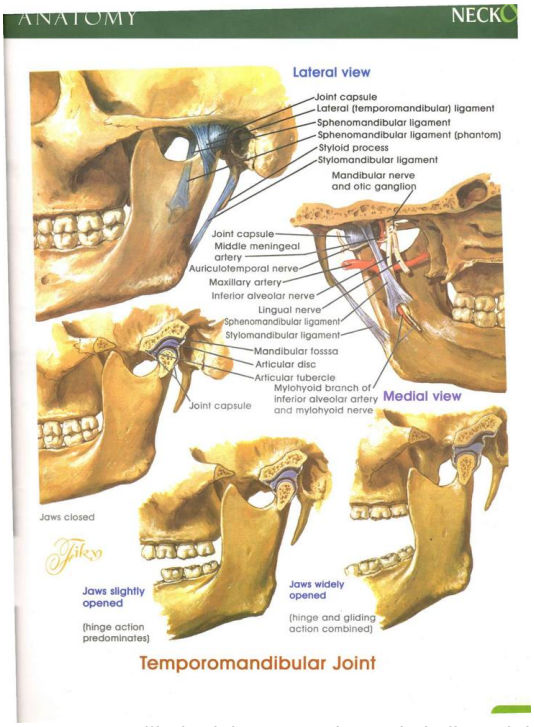



Mandible, medial view: pterygoid and constrictor attachments. Gray's Atlas of Anatomy, 2nd ed., p. 489

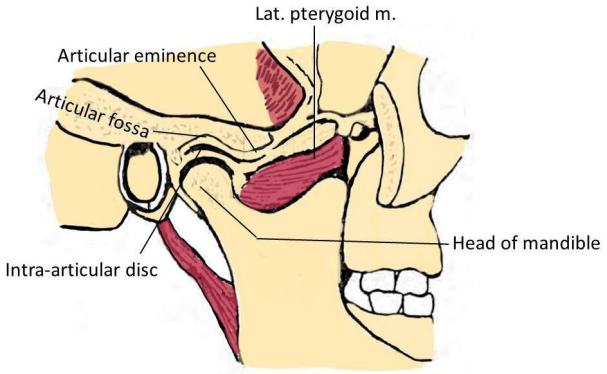
## Condyle, coronoid and notch

- **Head of the condyle:** articulates through the disc with the mandibular fossa and articular tubercle.
- **Pterygoid fovea:** a pit on the front of the neck, for the **lateral pterygoid**.
- **Coronoid process: temporalis**, on its tip and medial surface, extending down to the retromolar fossa.
- **Mandibular notch:** transmits the **masseteric nerve and vessels**.
- **Medial to the neck:** the auriculotemporal nerve and maxillary artery.
- The joint itself is covered under **Infratemporal fossa & TMJ**.

**Clinical pearl:** Anterior dislocation occurs when the condyle passes in front of the articular tubercle.




Temporomandibular joint. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 12

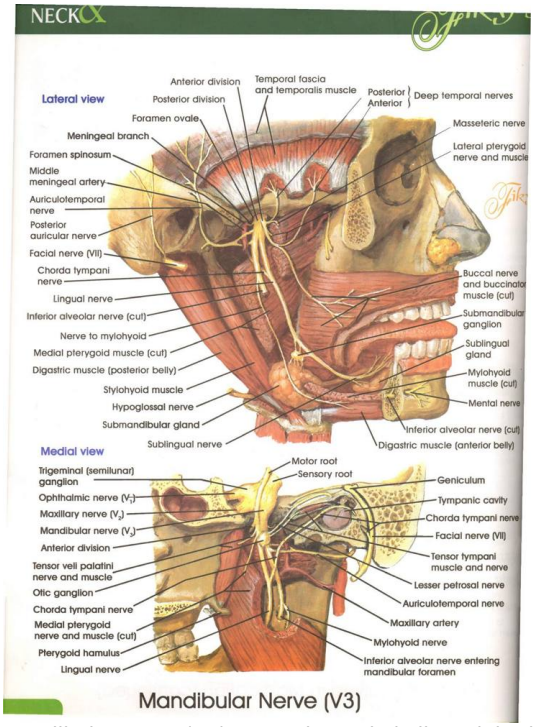



Temporomandibular joint: articular disc and lateral pterygoid. Dr. Sameh Doss handout, p. 199

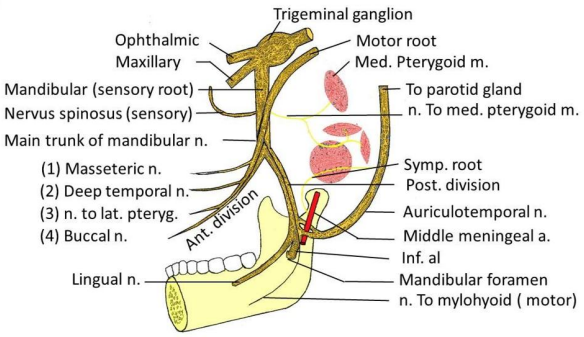
## Nerves related to the mandible

- **Inferior alveolar nerve:** enters the mandibular foramen, runs in the canal, supplies the lower teeth, then divides into **mental** and **incisive** branches.
- **Nerve to mylohyoid:** leaves before the foramen; supplies mylohyoid and the anterior belly of digastric.
- **Lingual nerve:** medial to the ramus, then beside the third molar, into the floor of the mouth.
- **Marginal mandibular branch of VII:** runs near the lower border, superficial to the facial vessels. **Auriculotemporal nerve:** behind the neck.

**Clinical pearl:** Lower lip numbness after mandibular surgery or fracture suggests inferior alveolar or mental nerve injury.




Mandibular nerve (V3). Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 21

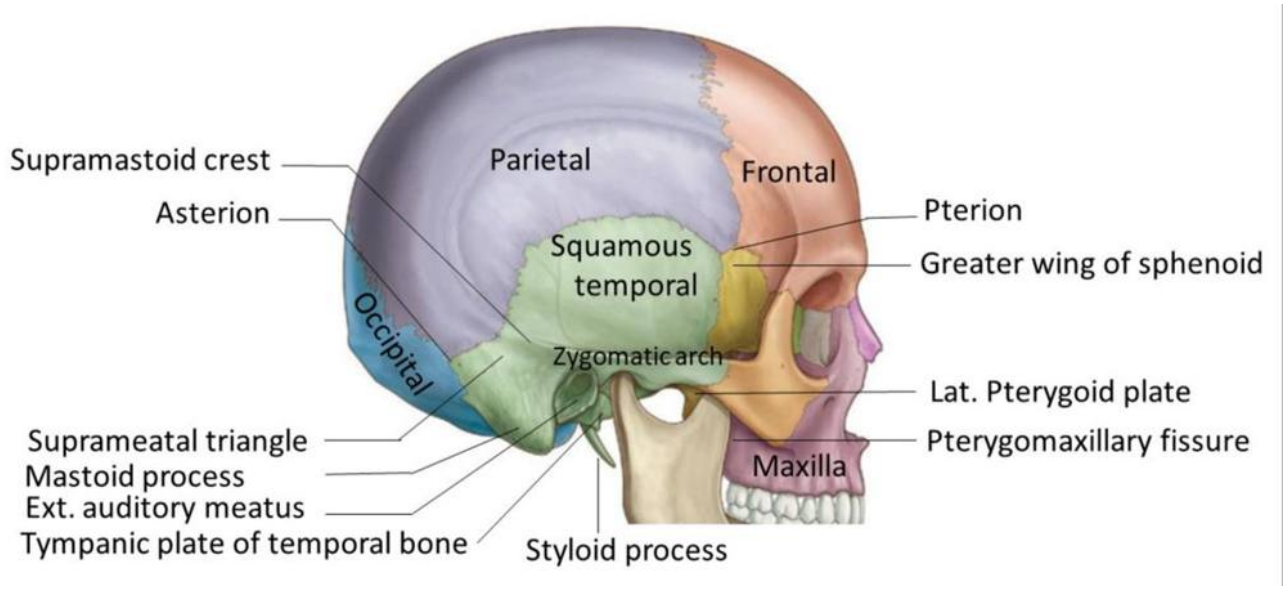



Mandibular nerve: trunk, divisions and branches. Dr. Sameh Doss handout, p. 55

## Age changes and clinical points

- **Angle:** obtuse in the infant, about **110–120°** in the adult, and obtuse again in the edentulous elderly (about 140°).
- **Mental foramen:** near the lower border in the infant, midway in the dentate adult, and near the upper edge after alveolar bone is lost.
- **Symphysis:** the two halves fuse in about the first year of life.
- **Fracture sites:** condylar neck, angle, body and parasymphysis are the common ones, often in combination.

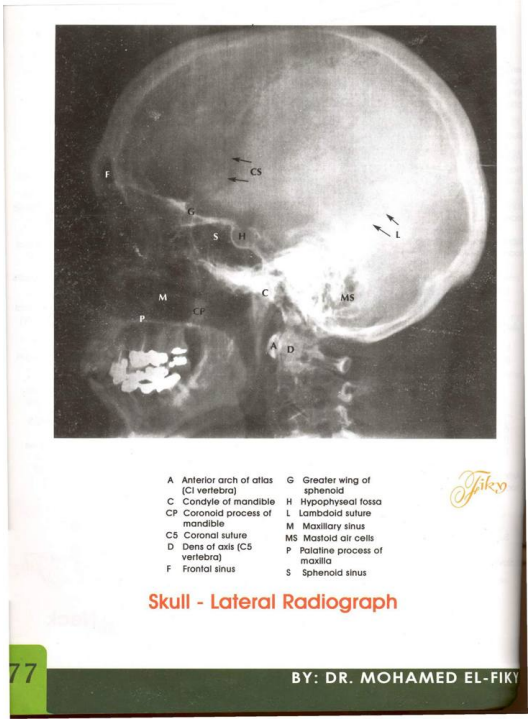
**Clinical pearl:** Retromolar trigone cancer lies directly on the mandible, so it can invade bone early.



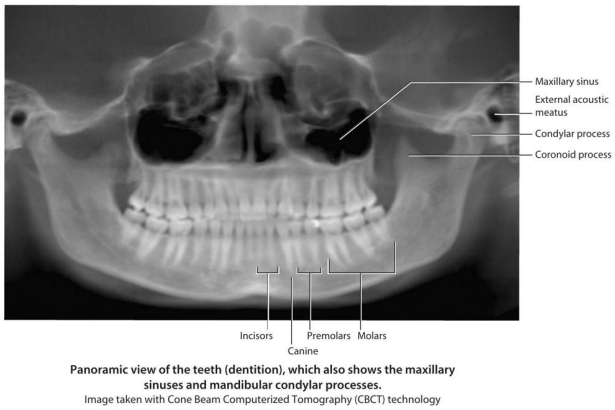

Lateral skull: pterion, asterion and the infratemporal surface. Dr. Sameh Doss handout, p. 9

## On imaging

- **Panoramic radiograph:** shows the teeth, mandibular canal, condyles and the floor of the maxillary sinus.
- **Lateral skull radiograph:** shows the condyle and coronoid process.



Skull, lateral radiograph. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 77



Panoramic radiograph: teeth and maxillary sinuses. Gray's Atlas of Anatomy, 2nd ed., p. 574

## In theatre

- **Mandibulotomy:** a paramedian cut, usually between the lateral incisor and canine, avoids dividing the muscles of the mental spines and spares the mental nerve.
- **Neck incisions:** stay below the lower border to protect the marginal mandibular nerve. **Third molar region:** protect the lingual nerve.

## Muscle attachments on the mandible

| Site | Muscle or ligament | Nerve |
|---|---|---|
| Lateral ramus and angle | Masseter | V3 |
| Coronoid process | Temporalis | V3 |
| Pterygoid fovea | Lateral pterygoid | V3 |
| Medial angle | Medial pterygoid | V3 |
| Mylohyoid line | Mylohyoid | Nerve to mylohyoid (V3) |
| Digastric fossa | Anterior belly of digastric | Nerve to mylohyoid (V3) |
| Superior mental spine | Genioglossus | XII |
| Inferior mental spine | Geniohyoid | C1 via XII |
| Lingula | Sphenomandibular ligament | Not applicable |

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